EPL strategy robustness — exploratory companion draft

October 14, 2026 | Separate extension of the frozen EPL study and the 172-row trading-metrics supplement. All analyses reuse already inspected history; no model, threshold, feature, outcome or selected trade is changed.

Saved analysis run: 2026-10-15T02:46:39.840013+00:00 UTC; October 14, 2026 at 10:46 PM EDT in New York. The source UTC timestamp and local draft date are recorded separately.

The stress table contains 1,422 rows: 158 original strategy/scope records under nine declared assumptions. The original 14 tuning-diagnostic rows are outside this stress extension. All 16 first locked-test policies and 22 pooled ablation variants remain visible, including cash policies. Every scenario retains hit ratio, expected gain/loss, frequency, realised returns and risk in the workbook.

These are hypothetical stress assumptions, not observed bookmaker fees or attainable prices. With original stake s, odds O, fee rate c on every wager and winning-profit haircut h, winning net payoff is $s\cdot[(O-1)(1-h)-c]$, and losing net payoff is $-s\cdot(1+c)$. The haircut applies to winning net payout before fees, excluding returned stake; it is not a haircut of full decimal odds.

Original selections, probabilities and allocated stakes stay fixed. There is no requalification, resizing or newly simulated Kelly sizing path. Fees are 0%, 0.25%, 0.5%, 1% or 2% of original stakes; payout haircuts are 1%, 2.5% or 5%; the combined stress uses a 1% fee plus a 2.5% haircut. Hit ratio and recorded trade frequency therefore remain unchanged by construction. Profitable-trade share remains separately defined from outcome hit ratio.

Model-implied expected gain/loss are probability-weighted positive/negative outcome cashflow components at those recorded stakes. They are not unconditional forecasts of terminal wealth. ROI divides stressed profit by original wager turnover; the separate cash-outlay ROI includes hypothetical fees in its denominator. The same 5% qualification rule already embeds positive model-implied value, so large expected returns are not independent evidence of an edge.

Funding is checked before date settlement: original date stakes plus fees must fit the stressed date-start cash. A failure leaves only an unfundable shadow cashflow calculation, marked †. Conventional max drawdown is then unavailable; shadow drawdown can exceed 100%. Each pooled season resets capital independently. No shadow path is presented as a feasible trading strategy or a fresh Kelly replay.

Reported stress intervals, where available, condition on the recorded positions and use the same whole-date bootstrap draws across scenarios, stratified by reset episode. They omit fitting, selection, market execution and multiplicity uncertainty. Bin, deletion and concentration summaries are descriptive diagnostics; they do not create a new holdout or justify a newly selected tail filter.

First locked test: all 16 policies under every stress

Cells show net profit / original wager turnover (%). Fee columns charge every original wager; payout columns reduce winning net payouts. A dash is undefined, including cash-policy ROI. † means the recorded exposure cannot be funded under that scenario.

Table R1. Complete 2025/26 locked-test policy grid; unchanged selections/stakes. These are assumptions applied to history, not actual provider fee schedules.

Strategy	Base	Fee 0.25%	Fee 0.5%	Fee 1%	Fee 2%	Payout −1%	Payout −2.5%	Payout −5%	Fee 1% + payout −2.5%
Accuracy / Fixed 5% Best book / Fixed	−2.25%	−2.50%	−2.75%	−3.25%	−4.25%	−2.96%	−4.02%	−5.80%	−5.02%
Accuracy / Fixed 5% Best book / Quarter Kelly	−7.01%	−7.26%	−7.51%	−8.01%	−9.01%	−7.69%	−8.69%	−10.38%	−9.69%
Accuracy / Fixed 5% Bet365 / Fixed	−5.12%	−5.37%	−5.62%	−6.12%	−7.12%	−5.80%	−6.82%	−8.52%	−7.82%
Accuracy / Fixed 5% Bet365 / Quarter Kelly	−8.57%	−8.82%	−9.07%	−9.57%	−10.57%	−9.22%	−10.20%	−11.83%	−11.20%
Calibration / Fixed 5% Best book / Fixed	0.41%	0.16%	−0.09%	−0.59%	−1.59%	−0.32%	−1.42%	−3.26%	−2.42%
Calibration / Fixed 5% Best book / Quarter Kelly	−2.59%	−2.84%	−3.09%	−3.59%	−4.59%	−3.29%	−4.35%	−6.12%	−5.35%
Calibration / Fixed 5% Bet365 / Fixed	−0.75%	−1.00%	−1.25%	−1.75%	−2.75%	−1.46%	−2.54%	−4.33%	−3.54%
Calibration / Fixed 5% Bet365 / Quarter Kelly	−3.09%	−3.34%	−3.59%	−4.09%	−5.09%	−3.78%	−4.82%	−6.56%	−5.82%
Accuracy / Primary Best book / Fixed	—	—	—	—	—	—	—	—	—
Accuracy / Primary Best book / Quarter Kelly	—	—	—	—	—	—	—	—	—
Accuracy / Primary Bet365 / Fixed	—	—	—	—	—	—	—	—	—
Accuracy / Primary Bet365 / Quarter Kelly	—	—	—	—	—	—	—	—	—
Calibration / Primary Best book / Fixed	—	—	—	—	—	—	—	—	—
Calibration / Primary Best book / Quarter Kelly	—	—	—	—	—	—	—	—	—
Calibration / Primary Bet365 / Fixed	—	—	—	—	—	—	—	—	—
Calibration / Primary Bet365 / Quarter Kelly	—	—	—	—	—	—	—	—	—

Pooled ablation: all 22 variants under every stress

Four exploratory outer seasons, 2022/23–2025/26; independently funded resets. Pooled profit and turnover add across seasons. † marks a failure in at least one episode; its value is a shadow cashflow return.

For 11 of 11 positive baseline point returns, the conditional 95% ROI interval includes zero. Full scenario intervals remain in the workbook; these overlapping historical comparisons do not establish a newly selected winning strategy.

Table R2. Complete pooled variant/staking grid. No displayed scenario or strategy is selected for deployment from these results.

Strategy	Base	Fee 0.25%	Fee 0.5%	Fee 1%	Fee 2%	Payout −1%	Payout −2.5%	Payout −5%	Fee 1% + payout −2.5%
Class frequency / Fixed	−4.85%	−5.10%	−5.35%	−5.85%	−6.85%	−5.58%	−6.66%	−8.46%	−7.66%
Class frequency / Quarter Kelly	−8.13%	−8.38%	−8.63%	−9.13%	−10.13%	−8.83%	−9.87%	−11.62%	−10.87%
Raw market / Fixed	−31.59%	−31.84%	−32.09%	−32.59%	−33.59%	−32.17%	−33.06%	−34.53%	−34.06%
Raw market / Quarter Kelly	−27.27%	−27.52%	−27.77%	−28.27%	−29.27%	−27.88%	−28.79%	−30.31%	−29.79%
Logistic: Combined / Fixed	0.14%	−0.11%	−0.36%	−0.86%	−1.86%	−0.57%	−1.63%	−3.39%	−2.63%
Logistic: Combined / Quarter Kelly	−3.47%	−3.72%	−3.97%	−4.47%	−5.47%	−4.13%	−5.12%	−6.77%	−6.12%
Logistic: Team / Fixed	1.59%	1.34%	1.09%	0.59%	−0.41%	0.87%	−0.21%	−2.01%	−1.21%
Logistic: Team / Quarter Kelly	−3.13%	−3.38%	−3.63%	−4.13%	−5.13%	−3.79%	−4.78%	−6.44%	−5.78%
Logistic: Team+players / Fixed	−1.73%	−1.98%	−2.23%	−2.73%	−3.73%	−2.43%	−3.47%	−5.20%	−4.47%
Logistic: Team+players / Quarter Kelly	−5.97%	−6.22%	−6.47%	−6.97%	−7.97%	−6.62%	−7.58%	−9.19%	−8.58%
Logistic: Team+xG / Fixed	7.01%	6.76%	6.51%	6.01%	5.01%	6.24%	5.09%	3.18%	4.09%
Logistic: Team+xG / Quarter Kelly	−0.59%	−0.84%	−1.09%	−1.59%	−2.59%	−1.27%	−2.30%	−4.00%	−3.30%
Market residual: Combined / Fixed	0.76%	0.51%	0.26%	−0.24%	−1.24%	0.06%	−0.99%	−2.74%	−1.99%
Market residual: Combined / Quarter Kelly	7.19%	6.94%	6.69%	6.19%	5.19%	6.48%	5.42%	3.64%	4.42%
Market residual: Team / Fixed	0.91%	0.66%	0.41%	−0.09%	−1.09%	0.20%	−0.85%	−2.62%	−1.85%
Market residual: Team / Quarter Kelly	1.84%	1.59%	1.34%	0.84%	−0.16%	1.18%	0.18%	−1.48%	−0.82%
Market residual: Team+players / Fixed	2.03%	1.78%	1.53%	1.03%	0.03%	1.32%	0.26%	−1.51%	−0.74%
Market residual: Team+players / Quarter Kelly	5.74%	5.49%	5.24%	4.74%	3.74%	5.04%	3.99%	2.25%	2.99%
Market residual: Team+xG / Fixed	0.42%	0.17%	−0.08%	−0.58%	−1.58%	−0.28%	−1.33%	−3.08%	−2.33%
Market residual: Team+xG / Quarter Kelly	1.46%	1.21%	0.96%	0.46%	−0.54%	0.79%	−0.21%	−1.88%	−1.21%
Temperature market / Fixed	−28.62%	−28.87%	−29.12%	−29.62%	−30.62%	−29.07%	−29.76%	−30.91%	−30.76%
Temperature market / Quarter Kelly	−16.22%	−16.47%	−16.72%	−17.22%	−18.22%	−16.61%	−17.21%	−18.19%	−18.21%

First locked test: combined-stress trade metrics

Combined assumption: 1% of each original stake plus a 2.5% haircut of winning profit. All money columns are units per trade; average +gain/−loss are realised conditional net-P&L magnitudes. Hit and frequency are percentages.

Table R3. All 16 policies. Funding cell is Yes/conventional drawdown, or No/unavailable with separately labelled shadow drawdown. Totals, probability-weighted net expectancy and all other scenario metrics remain in the workbook.

Strategy	Trades	Hit %	Trade freq. %	Avg. + gain	Avg. − loss size	Expected gain/trade	Expected loss/trade	Realised net/trade	Fundable / max. DD
Accuracy / Fixed 5% Best book / Fixed	371	26.68%	99.20%	25.87	10.10	11.41	6.38	−0.50	Yes / 30.23%
Accuracy / Fixed 5% Best book / Quarter Kelly	371	26.68%	99.20%	35.48	14.83	16.99	9.14	−1.40	Yes / 58.72%
Accuracy / Fixed 5% Bet365 / Fixed	365	26.85%	97.59%	24.61	10.10	10.56	6.31	−0.78	Yes / 33.59%
Accuracy / Fixed 5% Bet365 / Quarter Kelly	365	26.85%	97.59%	32.71	14.13	15.01	8.63	−1.55	Yes / 61.19%
Calibration / Fixed 5% Best book / Fixed	333	27.03%	89.04%	26.37	10.10	12.06	6.38	−0.24	Yes / 33.27%
Calibration / Fixed 5% Best book / Quarter Kelly	333	27.03%	89.04%	45.39	18.12	22.31	11.19	−0.96	Yes / 63.19%
Calibration / Fixed 5% Bet365 / Fixed	314	27.71%	83.96%	25.08	10.10	11.28	6.35	−0.35	Yes / 34.18%
Calibration / Fixed 5% Bet365 / Quarter Kelly	314	27.71%	83.96%	42.83	17.83	20.43	10.95	−1.02	Yes / 60.39%
Accuracy / Primary Best book / Fixed	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Accuracy / Primary Best book / Quarter Kelly	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Accuracy / Primary Bet365 / Fixed	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Accuracy / Primary Bet365 / Quarter Kelly	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Calibration / Primary Best book / Fixed	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Calibration / Primary Best book / Quarter Kelly	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Calibration / Primary Bet365 / Fixed	0	—	0.00%	—	—	—	—	—	Yes / 0.00%
Calibration / Primary Bet365 / Quarter Kelly	0	—	0.00%	—	—	—	—	—	Yes / 0.00%

Pooled ablation: combined-stress trade metrics

Same combined assumption and columns as R3. Pooled drawdown describes the worst separate episode; it is not a continuous bankroll path. Counterfactual recorded activity is not a claim of feasible executed activity.

Table R4. Every pooled variant/staking rule. Missing per-trade means for no-trade policies remain unavailable, never invented zero payoffs.

Strategy	Trades	Hit %	Trade freq. %	Avg. + gain	Avg. − loss size	Expected gain/trade	Expected loss/trade	Realised net/trade	Fundable / max. DD
Class frequency / Fixed	1,447	23.01%	98.44%	30.46	10.10	16.92	6.18	−0.77	Yes / 54.03%
Class frequency / Quarter Kelly	1,447	23.01%	98.44%	43.73	15.17	25.58	9.16	−1.61	Yes / 82.28%
Raw market / Fixed	263	9.51%	17.89%	60.32	10.10	9.65	8.73	−3.41	Yes / 40.77%
Raw market / Quarter Kelly	263	9.51%	17.89%	22.46	3.54	3.46	3.09	−1.07	Yes / 14.23%
Logistic: Combined / Fixed	1,285	29.49%	87.41%	23.25	10.10	9.25	6.53	−0.26	Yes / 70.63%
Logistic: Combined / Quarter Kelly	1,285	29.49%	87.41%	30.24	13.86	12.91	8.68	−0.85	Yes / 70.65%
Logistic: Team / Fixed	1,241	29.57%	84.42%	23.65	10.10	9.35	6.55	−0.12	Yes / 30.68%
Logistic: Team / Quarter Kelly	1,241	29.57%	84.42%	30.47	13.95	13.21	8.73	−0.81	Yes / 54.34%
Logistic: Team+players / Fixed	1,274	28.81%	86.67%	23.41	10.10	9.37	6.54	−0.45	Yes / 42.64%
Logistic: Team+players / Quarter Kelly	1,274	28.81%	86.67%	29.31	13.49	12.75	8.43	−1.16	Yes / 63.92%
Logistic: Team+xG / Fixed	1,236	30.42%	84.08%	24.45	10.10	9.23	6.53	0.41	Yes / 26.38%
Logistic: Team+xG / Quarter Kelly	1,236	30.42%	84.08%	32.77	15.04	13.90	9.33	−0.50	Yes / 56.44%
Market residual: Combined / Fixed	1,186	30.69%	80.68%	22.16	10.10	8.33	6.70	−0.20	Yes / 49.78%
Market residual: Combined / Quarter Kelly	1,186	30.69%	80.68%	33.87	14.04	12.05	9.33	0.67	Yes / 39.97%
Market residual: Team / Fixed	1,048	30.34%	71.29%	22.57	10.10	8.24	6.77	−0.19	Yes / 39.87%
Market residual: Team / Quarter Kelly	1,048	30.34%	71.29%	27.50	12.13	10.20	8.06	−0.11	Yes / 29.06%
Market residual: Team+players / Fixed	1,147	31.04%	78.03%	22.20	10.10	8.28	6.70	−0.07	Yes / 41.04%
Market residual: Team+players / Quarter Kelly	1,147	31.04%	78.03%	31.99	13.76	11.64	9.09	0.44	Yes / 39.54%
Market residual: Team+xG / Fixed	1,098	30.33%	74.69%	22.44	10.10	8.26	6.73	−0.23	Yes / 25.17%
Market residual: Team+xG / Quarter Kelly	1,098	30.33%	74.69%	28.90	12.82	10.73	8.39	−0.16	Yes / 27.67%
Temperature market / Fixed	216	25.46%	14.69%	17.48	10.10	7.58	6.88	−3.08	Yes / 58.61%
Temperature market / Quarter Kelly	216	25.46%	14.69%	12.71	6.42	4.59	4.12	−1.55	Yes / 21.78%

Selected-bet tails: expected and realised are different

Descriptive illustration: calibration/best-book/fixed 5% policy, with 333 recorded bets in the first test. In saved bins at odds ≥ 10, 20 bets win 0 times while their mean forecast probability is 20.29%. This is a small retrospective tail, not evidence for a stable exclusion rule.

Across these selected bets, mean forecast p is 36.83% versus hit ratio 27.03%; binary Brier is 0.1988, log loss 0.5837, and ten-bin ECE 0.0981. Model-implied ROI is 60.86% versus realised 0.41%. These conditional binary diagnostics differ from full-match three-outcome scores.

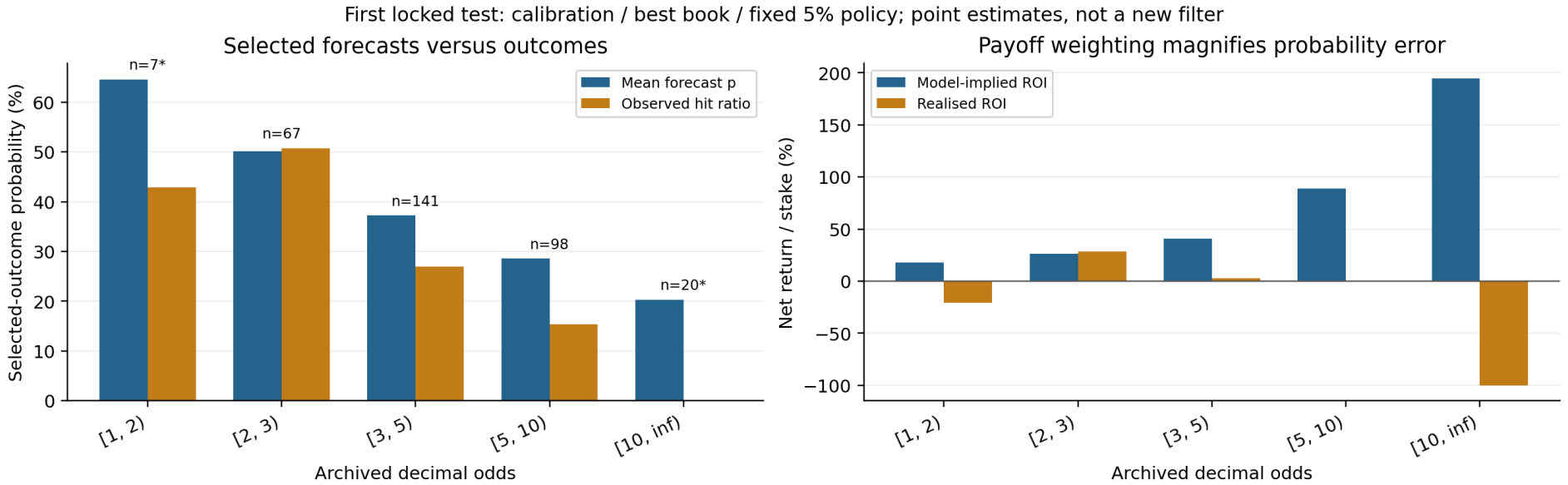


*Figure R1. Saved selected-bet bins; point estimates. * denotes fewer than 30 trades, with no implied precision. High-odds payoff weights can enlarge modest probability errors; empty bins remain unobserved.*

Table R5. Original baseline positions only. For a binary recorded wager, expected minus realised net P&L equals stake×odds×(p−outcome); summing gives an arithmetic decomposition, not a causal explanation or prospective result.

Odds bin	Trades	Wins	Mean p	Hit ratio	Expected ROI	Realised ROI
[1, 2)	7	3	64.59%	42.86%	17.91%	−20.57%
[2, 3)	67	34	50.11%	50.75%	26.53%	28.73%
[3, 5)	141	38	37.24%	26.95%	40.81%	3.02%
[5, 10)	98	15	28.57%	15.31%	88.98%	−0.71%
[10, inf)	20	0	20.29%	0.00%	194.42%	−100.00%

Date concentration and season fragility

Top-date shares divide the largest positive date P&L sums by total positive date P&L, not by small overall net profit. Deletion ranges remove each entire episode/date or season while keeping all other recorded bets/stakes fixed; they are not refitted strategies or confidence intervals.

Table R6. All 22 pooled baseline variants. Profitable seasons are observed signs. Overlapping strategies reuse fixtures; rows are not independent replications. Detailed date influence, season deletions and selected-bet scores remain in the workbook.

Strategy	Base ROI	Top 1 date positive share	Top 5 dates positive share	Delete-one-date ROI range	Profitable seasons	Delete-one-season ROI range
Class frequency / Fixed	−4.85%	3.83%	17.26%	[−6.39%, −4.26%]	1/4	[−7.71%, −2.73%]
Class frequency / Quarter Kelly	−8.13%	4.16%	16.86%	[−9.78%, −7.35%]	0/4	[−11.46%, −6.15%]
Raw market / Fixed	−31.59%	13.65%	43.49%	[−39.05%, −30.53%]	0/4	[−39.83%, −22.67%]
Raw market / Quarter Kelly	−27.27%	9.29%	36.40%	[−32.69%, −26.23%]	0/4	[−36.31%, −20.41%]
Logistic: Combined / Fixed	0.14%	2.89%	12.05%	[−1.03%, 0.61%]	3/4	[−4.55%, 6.77%]
Logistic: Combined / Quarter Kelly	−3.47%	2.29%	9.76%	[−4.32%, −2.94%]	1/4	[−7.36%, −0.12%]
Logistic: Team / Fixed	1.59%	4.87%	13.53%	[−0.39%, 2.24%]	3/4	[0.10%, 4.33%]
Logistic: Team / Quarter Kelly	−3.13%	3.12%	12.29%	[−4.26%, −2.58%]	0/4	[−4.09%, −1.00%]
Logistic: Team+players / Fixed	−1.73%	2.69%	11.35%	[−2.75%, −1.13%]	2/4	[−4.09%, 1.02%]
Logistic: Team+players / Quarter Kelly	−5.97%	2.37%	10.85%	[−6.82%, −5.31%]	1/4	[−9.44%, −4.22%]
Logistic: Team+xG / Fixed	7.01%	4.37%	13.05%	[5.03%, 7.53%]	4/4	[4.76%, 9.05%]
Logistic: Team+xG / Quarter Kelly	−0.59%	3.25%	12.22%	[−1.86%, 0.04%]	2/4	[−4.12%, 1.85%]
Market residual: Combined / Fixed	0.76%	2.86%	11.96%	[−0.39%, 1.36%]	2/4	[−5.36%, 4.56%]
Market residual: Combined / Quarter Kelly	7.19%	2.89%	11.71%	[6.02%, 7.91%]	2/4	[−0.60%, 11.06%]
Market residual: Team / Fixed	0.91%	4.90%	14.89%	[−1.18%, 1.49%]	2/4	[−2.07%, 4.24%]
Market residual: Team / Quarter Kelly	1.84%	2.43%	10.67%	[0.88%, 2.50%]	2/4	[−0.37%, 3.70%]
Market residual: Team+players / Fixed	2.03%	3.56%	12.49%	[0.55%, 2.66%]	2/4	[−3.64%, 5.38%]
Market residual: Team+players / Quarter Kelly	5.74%	2.80%	11.78%	[4.61%, 6.34%]	2/4	[−1.71%, 8.71%]
Market residual: Team+xG / Fixed	0.42%	3.42%	12.48%	[−0.95%, 0.92%]	2/4	[−0.73%, 1.55%]
Market residual: Team+xG / Quarter Kelly	1.46%	2.41%	9.90%	[0.54%, 2.11%]	4/4	[1.10%, 1.80%]
Temperature market / Fixed	−28.62%	16.13%	48.99%	[−34.87%, −27.61%]	1/4	[−41.94%, −3.85%]
Temperature market / Quarter Kelly	−16.22%	6.42%	25.51%	[−18.22%, −14.52%]	0/4	[−38.39%, −7.18%]

Coverage, empirical cost buffers and reproducibility

The complete workbook contains all 1,422 scenario rows and 158 empirical break-even rows; 24 baseline scopes hold no trades. It also retains 2,016 per-scenario/episode funding summaries, diagnostic availability, probability/odds/edge bins, selected-bet scores, date influence and season fragility. Across all scopes/scenarios, 0 rows have unfundable recorded exposure; these are not independent failure observations. All nine mild assumptions passed the funding check in this saved run; this does not verify price execution or future funding.

Historical break-even fee equals baseline net profit / turnover; the winning-profit haircut buffer equals baseline net profit / total winning net gains. These signed arithmetic buffers use observed payoffs and cannot be treated as forecasts of allowable future costs. A negative buffer means the original historical return was already below zero.

For the first-test calibration/best-book/fixed policy, historical ROI is 0.41%. A hypothetical 0.5% charge on every original wager changes it to −0.09% without changing its selected outcomes or recorded frequency. This quantifies sensitivity under one stated assumption; it does not identify an actual fee or an executable advantage.

Selected-outcome calibration is conditional on which bets were chosen. Bin hit ratios differ from full-match H/D/A accuracy and multiclass calibration. Model probability gaps and financial expectation gaps have different weighting; odds and stakes amplify some errors. Sparse tails, repeated clubs, overlapping strategies and already inspected outcomes restrict inference. The present-day club cohort retains its original survivor and source-coverage limits. No new model or betting filter is recommended from this extension.

A dash denotes undefined/unavailable, and genuine infinity remains explicit. A no-trade policy has zero cash totals and activity when opportunities exist, while hit ratio, per-trade means and ROI remain undefined. Failed funding does not erase a recorded outcome, but it prevents interpreting its shadow path as a feasible wager sequence.

This companion is separate from the seven-page draft and the 172-row supplement. Its manifest hashes every new analytic input and the unchanged original study/report sources. Rebuild with python -m epl_robustness.report --results PATH. Rendering does not collect data, fit models, change selection or overwrite prior results. Full-precision source tables remain beside the companion; display CSVs use the published rounding.